



PRODUCT REQUIREMENTS and SPECIFICATION DOCUMENT (PRD)

AI Performance & QA Monitoring System (MVP) – PHASE 1

Automatically evaluate interactions, monitor performance metrics, and generate actionable insights in real time—enabling consistent quality, reduced manual QA effort, and data-driven performance management across operations.

Category: AI Customer Support

Type: MVP Product (Integration-Based Solution)

Purpose: MVP Definition and Estimation Input

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1. TARGET INDUSTRY / MARKET

This section defines the primary industries where the product is most applicable for initial deployment and commercialization.

Primary Target Industry (MVP Focus)

BPO / Call Centers (Customer Support & Moderation Operations)

- High volume of customer interactions (calls, chat, email) requiring quality evaluation and performance tracking
- Strong need for consistent QA scoring and standardized evaluation frameworks
- Limited QA coverage (sampling-based), creating gaps in performance visibility
- Demand for real-time performance monitoring and coaching insights
- Operational pressure to reduce manual QA effort and improve efficiency
- Need for compliance monitoring and service quality assurance

Secondary Target Industries

- **SaaS & Tech Companies (Customer Support / Success Teams)** (order reviews, product uploads, user comments)
 - Monitor support interactions for **quality, resolution effectiveness, and customer experience**
 - Identify trends impacting **retention and satisfaction (CSAT proxy)**
 - Enable **data-driven coaching and performance management**

E-commerce & Marketplaces (Customer Support Operations)

1. Evaluate support interactions across order handling, complaints, and returns
2. Ensure consistent service quality during high-volume periods
3. Improve customer experience and operational efficiency

Shared Services / Enterprise Support Functions

- Monitor internal support teams (IT helpdesk, HR support)
- Standardize service quality across departments
- Provide visibility into team performance and operational efficiency

Positioning

Designed as an **industry-agnostic QA and performance intelligence platform**, adaptable to any organization managing high volumes of customer or operational interactions.

Prioritized for industries with:

- High interaction volume (voice, chat, email)
- Need for **quality assurance and performance tracking at scale**
- Manual or inconsistent QA processes
- Strong focus on **customer experience, compliance, and operational efficiency**

4. PROBLEM STATEMENT

This section outlines key challenges in customer support and moderation operations, supported by industry data.

4.1 Content Volume is Outpacing Moderation Capacity (Global)

User-generated content across chat, email, and digital platforms continues to grow rapidly, while moderation remains largely manual.

- **60–80%** of interactions are repetitive or require basic moderation (McKinsey, Gartner)
- Human-led processes cannot scale efficiently with volume
- Delays increase exposure to harmful or inappropriate content

4.2 Rising Compliance and Brand Risk from Inconsistent Moderation (Global)

Organizations face increasing pressure to enforce policies consistently across all interactions.

- Manual moderation leads to inconsistent decisions and compliance gaps
- Lack of auditability increases regulatory and legal risk
- Unmoderated content directly impacts brand trust and reputation

4.3 Manual Moderation Model Limits Scalability in the Philippines (Local Context)

The Philippine BPO industry remains heavily reliant on human moderation.

- ~1.7 million workforce supporting CX operations (IBPAP)
- High attrition impacts quality and continuity
- Scaling requires continuous hiring, increasing operational cost

5. SOLUTION OVERVIEW

This section outlines how the proposed solution addresses the identified challenges in customer support operations, particularly in high-volume call center and logistics environments.

The solution introduces an AI-powered omnichannel customer support agent that automates and standardizes customer interactions across voice and chat, handling both inbound and outbound requests. It enables faster, more consistent, and scalable support operations while integrating seamlessly with Zoho CRM and existing workflows.

5.1 Core Capabilities of the Solution

The AI Performance & QA Monitoring System transforms traditional QA into a **scalable, AI-driven process** by automating interaction evaluation, performance monitoring, and insight generation.

At its core, the platform analyzes **100% of interactions** in real time and batch, detecting quality issues, compliance risks, and sentiment signals. It proactively flags high-risk cases, triggers alerts, and enables immediate action—shifting QA from reactive to proactive.

The system then converts these evaluations into **actionable performance insights**, generating operator scorecards, automated reports, and daily summaries that provide clear visibility into trends, gaps, and coaching opportunities.

To ensure accuracy and control, a **Human-in-the-Loop (HITL)** layer allows QA teams to validate results, refine scoring, and continuously improve the system. This is supported by **smart sampling automation** and a complete audit trail for compliance and traceability.

Phase 1 – MVP Core Capabilities

1. AI-Powered Interaction Quality Evaluation & Monitoring

Agents: Content Quality Checker, AI Message Rating, and Sentiment Analysis

- Evaluates interactions in **real-time and batch**
- Detects:
 - Tone issues (incl. emoji usage)
 - Policy violations
 - Low-quality responses
 - Sentiment signals (frustration, urgency)
- Automatically:
 - Flags high-risk interactions
 - Disables low-quality AI responses
 - Triggers alerts (e.g., Slack)
- Enables **proactive QA and risk detection**

2. AI-Driven Performance Evaluation & Reporting

Agents: Employee Performance Evaluation, Operator Rating Agent, and AI Message Rating

- Generates:
 - Operator-level evaluations (summary, justification, actions)
 - Scorecards and cutoff-based reports
 - Daily QA alert & summary reports (email)
- Tracks:
 - QA scores
 - Escalations
 - Performance trends
- Provides:
 - Actionable insights
 - Coaching recommendations
- Automates **scheduled and real-time reporting**

3. Smart QA Sampling & Audit Automation

Agent: Random Message Selector

- Automates unbiased sampling
- Ensures representative QA coverage
- Reduces QA prep effort

4. Human-in-the-Loop (HITL) Validation & Control

- QA analysts validate flagged interactions
- Override scores and decisions
- Improve model accuracy over time

Together, these capabilities create a unified system that enables organizations to:

- Evaluate **100% of interactions** with consistency and accuracy
- Reduce manual QA effort by **up to 60–70%**
- Gain real-time visibility into performance and quality trends
- Enable faster coaching, issue resolution, and decision-making
- Establish a scalable foundation for AI-driven operational excellence

5.2 Solution Pre-requisites

Phase 1 – MVP Pre-requisites

A. QA Framework & Evaluation Criteria

- Defined QA scorecards and evaluation rubrics (e.g., compliance, communication quality, resolution)
- Standardized SOPs and quality guidelines per campaign or account
- Clear scoring logic and weighting criteria
- Defined thresholds for:
 - Low-quality detection

- Compliance violations
- Escalation triggers

B. Interaction Data Availability

- Access to historical and/or real-time interaction data:
 - Chat transcripts
 - Email threads
 - Voice transcripts (or recordings with STT capability)
- Sample datasets for initial configuration and testing
- Data structured and labeled (if available) for improved accuracy

C. Basic Knowledge Base (QA Guidelines & Policies)

- Documented SOPs, policies, and QA standards
- Defined examples of:
 - Good vs. poor quality interactions
 - Compliance vs. non-compliance cases
- Content mapped to QA evaluation criteria

D. System & Integration Readiness

- Access to relevant platforms (depending on setup):
 - Helpdesk tools (e.g., Zendesk, Freshdesk)
 - Communication platforms (chat, email, voice systems)
- API access (read capability minimum)
- Ability to extract or stream interaction data

E. Agent & Performance Data (for Reporting)

- Operator/agent identifiers linked to interactions
- Basic performance tracking fields (e.g., timestamps, resolution status)
- Historical data for generating baseline reports and trends

F. HITL & QA Operations Setup

- Defined QA reviewers or supervisors for validation
- Workflow for:
 - Reviewing flagged interactions
 - Overriding AI scores
 - Providing feedback
- Alignment on QA review process and turnaround time

G. Reporting & Notification Setup

- Defined recipients for reports (e.g., supervisors, managers)
- Email or Slack configuration for alerts and summaries
- Agreement on reporting cadence (daily, weekly, cutoff-based)

6. USE CASES (PHASE 1 - MVP)

This section defines the end-to-end workflows, including responsibilities across AI agents, system components, and human agents (HITL) where applicable.

Use Case 1: Automated Interaction Quality Evaluation

Scenario: High-Volume Interaction QA Coverage (Post Interaction coverage)

A customer support team handles thousands of daily interactions across chat, email, and voice. Due to limited QA resources, only a small percentage of these interactions are reviewed, leaving most quality issues undetected.

Volume Profile (MVP Assumption)

- 100–500 interactions/day
- Channels: Chat, Email, Voice (transcripts)
- Target: 100% interaction evaluation
- Processing: Near real-time or batch (within minutes)

Current State

- Only 1–5% of interactions reviewed
- Manual, inconsistent QA

With the Solution

The system connects to existing platforms (CRM, helpdesk, communication tools) and captures interaction data via **APIs or data feeds**. Without changing any agent workflows, all interactions are automatically evaluated in the background using standardized QA scorecards.

As interactions are processed, the AI identifies quality issues, compliance gaps, and sentiment signals, generating scores and flags for each case. High-risk or low-confidence interactions are routed to QA reviewers, allowing teams to focus only on cases that require human judgment.

HITL

- The system automatically prioritizes **high-risk, low-confidence, or policy-sensitive interactions** for human review
- QA analysts are presented with:
 - Full interaction context
 - AI-generated scores and detected issues
 - Clear reason for flagging (e.g., compliance risk, sentiment, low confidence)
- Analysts can:
 - Validate or override scores
 - Add feedback and corrections

- All human inputs are captured and fed back into the system to **continuously improve scoring accuracy and QA criteria**

Operational Interactions

- Works in parallel (non-intrusive)
- Requires access to interaction data (API or export)
- Supports batch + near real-time processing

Outcome

- 100% QA coverage
- Reduced manual effort
- Improved consistency

Use Case 2: Real-Time QA Monitoring & Alerting

Scenario: Live Interaction Quality Risk Detection & Intervention

Quality issues during live interactions are often detected too late, after the customer experience has already been impacted.

Volume Profile (MVP Assumption)

- 50–200 concurrent interactions
- Real-time processing (<1–3 seconds latency)
- Continuous monitoring during active sessions

Current State

- QA is post-interaction
- No real-time visibility

With the Solution

The system integrates with live interaction streams (chat or voice transcripts) and monitors conversations in real time without interfering with ongoing operations. As messages are exchanged, the AI evaluates responses instantly and detects issues such as poor quality, compliance risks, or negative sentiment.

When a risk is identified, the system immediately flags the interaction and sends alerts through existing channels (e.g., Slack or email). This allows supervisors to stay informed and intervene when necessary, while the system continues to operate seamlessly in the background.

HITL: When critical issues are detected in real time, the system escalates only **high-priority interactions** to supervisors

- Supervisors receive:
 - Real-time alerts with conversation snapshot
 - Risk classification (e.g., compliance, negative sentiment)

- Suggested action (e.g., monitor, intervene, escalate)
- Supervisors can:
 - Intervene in the interaction if needed
 - Allow the interaction to proceed
 - Flag for post-interaction review
- The system learns from supervisor actions to **refine alert thresholds and reduce noise over time**

Operational Interactions

- Requires real-time data access or streaming
- Configurable thresholds for alerts
- Runs as a **parallel monitoring layer**

Outcome

- 100% interaction visibility (vs. traditional 1–5% QA sampling)
- 60–70% reduction in manual QA effort through automation
- <1–3 minutes issue detection time (near real-time for critical cases)
- 80–90% of high-risk interactions automatically flagged

Use Case 3: AI-Driven Performance Reporting & Coaching Insights

Scenario: Continuous Performance Monitoring & Supervisor Coaching Enablement

Supervisors rely on delayed, manual reports, limiting their ability to coach effectively and respond to performance issues.

Volume Profile (MVP Assumption)

- Covers **all evaluated interactions (1,000–5,000/day)**
- Report cadence:
 - Daily summaries
 - Weekly / cutoff reports
- Historical tracking: up to 30–60 days

Current State

- Manual reporting
- Delayed insights
- Reactive coaching

With the Solution

The system aggregates QA results and performance data directly from existing tools and evaluated interactions. Without requiring additional input from agents or supervisors, it automatically generates structured reports, including daily summaries, operator scorecards, and performance trends.

These insights are delivered via email, providing supervisors with clear visibility into team performance, recurring issues, and coaching opportunities. Because the system operates on top of existing data sources, reporting is continuous, automated, and non-disruptive.

HITL (Conditional Step with Sentiment Escalation)

- The system highlights **priority agents, performance gaps, and recommended actions** based on QA insights
- Supervisors are guided to:
 - Review flagged performance issues
 - Validate AI-generated insights and recommendations
- Supervisors can:
 - Confirm or adjust coaching priorities
 - Add contextual feedback (e.g., external factors, edge cases)
 - Initiate targeted coaching actions
 - Feedback is used to **refine performance evaluation logic and improve recommendation relevance over time**

Outcome

- **Real-time performance visibility**
 - 100% interaction coverage (vs. 1–5% traditional QA sampling)
 - Daily and near real-time reporting availability
- **Faster, data-driven coaching**
 - **30–50% reduction in time to identify underperforming agents**
 - **40–60% faster coaching cycle** (from issue detection to action)
- **Improved team performance**
 - **5–15% improvement in QA scores** within 1–3 months
 - **Reduced recurring quality issues by 20–40%** through targeted coaching

6.1 Use Cases Overview

Capability	Use Case 1: Automated QA Evaluation	Use Case 2: Real-Time Monitoring	Use Case 3: Performance Reporting & Insights
Primary Purpose	Evaluate interaction quality at scale	Detect and act on live risks	Provide performance visibility & coaching insights
Timing	Post-interaction / near real-time	During interaction (real-time)	After aggregation (daily / periodic)
Data Source	Interaction transcripts (chat, email, voice)	Live conversation streams	Aggregated QA results + historical data
Core Function	Scoring, compliance checks, quality analysis	Live monitoring, alerting, risk detection	Reporting, trend analysis, performance insights
Output	QA scores, flags, categorized issues	Real-time alerts, risk notifications	Scorecards, summaries, coaching insights
HITL Role	Validate flagged interactions and refine scoring	Intervene on critical interactions when needed	Review insights and take coaching actions
User Persona	QA Analysts	Supervisors / Team Leads	Managers / Supervisors
Operational Impact	Improves QA coverage and consistency	Prevents issues before escalation	Enables data-driven performance improvement
System Behavior	Runs in background (non-intrusive)	Runs in parallel with live interactions	Runs on aggregated data (non-intrusive)
Complexity Level	Low (easy MVP deployment)	Medium (requires real-time processing)	Low (reporting layer)
Operational Risk	Low	Medium	Low

7. USER FLOW (ROLE-BASED)

This section defines the end-to-end workflow of the AI Performance & QA Monitoring System, outlining responsibilities across system components, AI agents, and human reviewers (HITL). The flow operates as a **non-intrusive layer**, running in parallel with existing customer interaction systems.

7.1 Core Workflow (Applies to All Use Cases – Chat, Voice, Email)

Step	Actor	Role Description	AI Agent / Component	System Action	Output
1	Customer / Agent	Interaction occurs via existing platforms (chat, voice, email)	External Systems (CRM, Helpdesk, CCaaS)	Sends/receives messages or calls	Interaction generated
2	System	Captures interaction data (real-time or batch)	Data Ingestion Layer	Pulls data via API, webhook, or stream	Interaction data available
3	AI	Evaluates interaction quality	Content Quality Checker / AI Message Rating	Analyzes tone, compliance, quality	QA score + detected issues
4	AI	Detects sentiment and risk signals	Sentiment Analysis + AI Message Rating	Identifies negative sentiment, urgency, risks	Flags (risk / priority level)
5	AI	Prioritizes interactions for review	QA Evaluation Engine	Classifies interactions (normal vs flagged)	Review queue created
6	AI	Triggers alerts (if real-time or high-risk)	Alerting Engine	Sends notifications (Slack/email)	Alert generated
7	Human (QA / Supervisor)	Reviews flagged interactions (HITL)	HITL Control Layer	Validates or overrides AI outputs	Final QA decision
8	AI / System	Aggregates performance data	Performance Evaluation Engine	Computes agent-level metrics and trends	Performance insights
9	System	Generates reports and summaries	Reporting Engine (Operator Rating Agent)	Creates daily summaries, scorecards	Reports generated
10	Human (Supervisor)	Reviews insights and takes action	—	Conducts coaching / actions based on insights	Coaching actions initiated
11	System	Logs all evaluations and outcomes	Audit & Logging Layer	Stores QA scores, flags, decisions	Audit trail
12	System (Optional)	Continuous improvement loop	Learning / Feedback Engine	Updates models based on HITL feedback	Improved accuracy

7.2 Human-in-the-Loop (HITL) Control Framework

AI evaluates customer interactions at scale across voice, chat, and email, while **critical QA decisions, edge cases, and high-risk interactions are validated by human reviewers** through a built-in Human-in-the-Loop (HITL) control layer.

Configurable thresholds (depending on client goals) and conditions are applied to automatically **trigger escalation or human validation** based on defined criteria, including:

Confidence Thresholds

- Low confidence in **QA scoring or evaluation results**
- Ambiguous or unclear interactions impacting evaluation accuracy
- Multiple possible interpretations affecting scoring consistency

Sentiment Triggers (MVP Scope)

- Negative sentiment detected (e.g., frustration, complaints, urgency)
- Repeated issues or signs of customer dissatisfaction
- Escalation-related keywords (e.g., “agent”, “complaint”, “supervisor”)

Data Quality & Completeness Issues

- Missing or incomplete interaction data (e.g., partial transcripts)
- Inconsistent or unclear context affecting evaluation accuracy
- Insufficient data to confidently assess quality or compliance

Quality & Compliance Risks

- Potential compliance violations (e.g., missing required disclosures)
- Deviation from SOPs or QA standards
- High-risk interactions requiring human validation

Workflow Failures / Exceptions

1. AI unable to complete evaluation due to system or data limitations
2. Errors in data ingestion or processing
3. Unsupported or out-of-scope interaction types

Interaction Risk Signals

4. Repeated negative sentiment or escalation indicators
5. Customer explicitly requests escalation or supervisor
6. High-priority or VIP interaction flag (if available)

Custom Business Rules (Configurable)

- Client-defined QA thresholds and escalation conditions
- Priority handling for specific interaction types or risk levels
- SLA-based escalation triggers

8. INTERACTION LOGGING & BASIC REPORTING (MVP)

This section defines how interactions are captured, evaluated, and reported to support **quality assurance, performance monitoring, and continuous improvement**.

8.1 Overview

The system automatically logs and evaluates all customer interactions across chat, email, and voice (transcripts), operating as a **non-intrusive layer on top of existing systems**.

Each interaction is enriched with QA evaluation results, sentiment signals, and performance indicators to ensure full traceability and enable actionable reporting.

8.2 Logged Data Elements

The following data points are captured for each interaction:

- Interaction identifier (ID)
- Customer identifier (if available)
- Channel (chat, email, voice – inbound/outbound)
- Timestamp (start, end, duration)
- Agent / operator identifier
- Interaction transcript or content

QA & Evaluation Data:

- QA score (overall and/or category-level if applicable)
- Detected quality issues (e.g., tone, compliance, resolution gaps)
- Compliance flags (if any)
- Sentiment classification (e.g., positive, neutral, negative)
- Risk level / priority flag

Operational Data:

- Escalation status (flagged, reviewed, validated)
- Reviewer action (if HITL applied – validated/overridden)
- Confidence level of AI evaluation (optional for MVP)

8.3 Reporting Format (MVP – Email-Based)

Reporting is delivered via an automated **Daily QA Alert & Summary Report**, providing stakeholders with a structured and actionable view of quality and performance.

Delivery Details

- **Frequency:** Daily (end-of-day or configurable)
- **Format:** Email (structured summary and optional attachments)
- **Recipients:** QA managers, operations leads, supervisors

Report Contents (MVP)

1. QA Summary Metrics

- Total interactions evaluated
- QA coverage (%)
- Average QA score
- Escalation / flag rate

2. High-Risk & Flagged Interactions

- Count of:
 - Low-quality interactions
 - Compliance risks
 - Negative sentiment cases
- Highlight of critical cases for review

3. Performance Insights

- Top performing agents (based on QA scores)
- Agents requiring attention (low scores / high flags)
- Key recurring quality issues

4. AI Monitoring Insights

- Number of interactions flagged by AI
- Alert triggers (real-time or batch)
- Summary of detected trends

8.4 Sample Report Layout (Email)

Subject: **Daily QA Performance & Alert Summary – [Date]**

A. QA Summary Overview

- Total Interactions Evaluated: [X]
- QA Coverage: [100% / X%]
- Average QA Score: [X%]
- Flagged Interactions: [X%]
- High-Risk / Compliance Cases: [X]

B. Channel Breakdown

Channel	Volume	Avg QA Score	Flag Rate
Chat	[X]	[X%]	[X%]
Voice	[X]	[X%]	[X%]
Email	[X]	[X%]	[X%]

C. Top Quality Issues Identified

Issue Type	Count	%
Tone / Communication Issues	[X]	[X%]
SOP / Process Gaps	[X]	[X%]
Compliance Risks	[X]	[X%]
Incomplete Resolution	[X]	[X%]

D. Risk & Escalation Insights (HITL Triggers)

- Negative sentiment cases: [X%]
- Low-confidence evaluations: [X%]
- Compliance-related flags: [X%]
- Interactions reviewed by QA (HITL): [X]

E. Agent Performance Highlights

- Top Performers:
 - [Agent Name] – [QA Score]
 - [Agent Name] – [QA Score]
- Agents Requiring Attention:
 - [Agent Name] – [QA Score] (Key issue: [e.g., compliance gaps])
 - [Agent Name] – [QA Score] (Key issue: [e.g., tone inconsistency])

F. Key Insights & Observations

- [Recurring issue or trend identified]
- [Operational risk or anomaly]
- [Improvement opportunity or recommendation]

G. Sample Flagged Interactions (Optional)

- [Example: Low QA score due to incomplete resolution]
- [Example: Compliance issue detected]
- [Example: Negative sentiment escalation]

SAMPLE EMAIL MOCK UP

Subject: Daily QA Performance & Alert Summary – April 8, 2026

Hi Team,

Please find below the summary of QA performance and interaction quality insights for **April 8, 2026**.

A. QA Summary Overview

- Total Interactions Evaluated: 3,250
- QA Coverage: 100%
- Average QA Score: 88%
- Flagged Interactions: 812 (25%)
- High-Risk / Compliance Cases: 146

B. Channel Breakdown

Channel	Volume	Avg QA Score	Flag Rate
Chat	1,820	90%	22%
Voice	1,430	85%	29%

C. Top Quality Issues Identified

Issue Type	Count	%
Tone / Communication Issues	410	13%
SOP / Process Gaps	265	8%
Compliance Risks	146	4%
Incomplete Resolution	98	3%

D. Risk & Escalation Insights (HITL Triggers)

- Negative sentiment cases: 34%
- Low-confidence evaluations: 31%
- Compliance-related flags: 20%
- Interactions reviewed by QA (HITL): 812

E. Agent Performance Highlights

Top Performers:

- Agent A – 95%
- Agent B – 93%

Agents Requiring Attention:

- Agent D – 78% (Low compliance adherence)
- Agent F – 81% (Tone inconsistency)

F. Key Insights & Observations

- Increased negative sentiment observed for delayed shipments in NCR region
- Higher flag rates driven by missed SOP steps in address change scenarios
- Voice channel shows lower QA scores compared to chat, indicating coaching opportunity

G. Sample Flagged Interactions

- Low QA score due to incomplete resolution in refund scenario
- Compliance issue detected (missing required disclosure)
- Negative sentiment escalation requiring supervisor review

If you have any questions or would like a more detailed breakdown, feel free to reach out.

Best,

AI QA Monitoring System

9. AGENT SPECIFICATIONS

This section defines the core AI agents that power the MVP solution. The system operates as a **modular, multi-agent QA and performance monitoring architecture**, where each agent is responsible for evaluating interaction quality, detecting risks, generating insights, and enabling continuous improvement.

A built-in **Human-in-the-Loop (HITL)** layer ensures accuracy and control by enabling validation, escalation, and feedback for low-confidence, high-risk, or sensitive cases.

Agent	Description	Key Functions	AI & Orchestration Considerations
9.1 Content Quality Checker (CQC)	Evaluates interaction quality based on predefined QA standards and guidelines	• Analyze tone, structure, and communication quality • Detect SOP adherence and response consistency • Identify quality issues (tone, resolution gaps)	• Config-driven evaluation rules • Supports real-time and batch processing • Adaptable per client QA framework
9.2 AI Message Rating Agent	Performs real-time and post-interaction QA scoring and monitoring	• Score interactions using QA criteria	• Supports real-time monitoring via event streams

		• Detect low-quality responses and violations • Trigger alerts and disable low-quality AI responses (if applicable)	• Integrates with alerting systems (e.g., Slack) • Enables proactive QA detection
9.3 Sentiment Analysis Agent	Detects sentiment and behavioral signals to identify high-risk interactions	• Classify sentiment (positive, neutral, negative) • Detect escalation signals (frustration, urgency) • Flag high-priority interactions	• Works across chat and voice transcripts • Integrated with QA scoring and HITL triggers • Supports real-time and batch analysis
9.4 Random Message Selector	Automates unbiased sampling of interactions for QA audits	• Select representative interaction samples • Eliminate human bias in QA sampling • Prepare audit-ready datasets	• Configurable sampling logic • Integrates with QA workflows • Supports batch processing
9.5 Employee Performance Evaluation Agent	Generates structured performance evaluations and coaching insights	• Analyze QA results per agent • Generate summaries, justifications, and recommendations • Identify performance gaps	• Uses configurable evaluation criteria • Integrates with reporting workflows • Supports automated insights generation
9.6 Operator Rating Agent	Produces standardized reports and scorecards for operators	• Generate individual and cutoff-based performance reports • Track historical performance trends • Export reports (PDF / structured output)	• Supports scheduled report generation • Configurable time periods and filters • Enables audit and traceability
9.7 HITL Control & Validation Agent	Manages human validation, escalation, and feedback loop	• Route flagged interactions to QA reviewers • Enable score validation and overrides • Capture feedback for continuous improvement	• Central orchestration layer for escalation • Applies configurable thresholds (confidence, risk) • Improves model accuracy over time
9.8 Interaction Logging & Reporting Agent	Captures interaction data and generates QA reports	• Log interactions, QA scores, flags, and outcomes • Generate daily QA summaries and insights • Maintain audit trail	• Integrates with data sources and storage • Supports email-based reporting (MVP) • Enables traceability and compliance

9.1 INTEGRATION & DATA EXCHANGE

This capability defines how the solution integrates with external systems to **ingest interaction data, evaluate quality, and generate insights**, without disrupting existing customer support workflows.

The system operates as a **non-intrusive overlay layer**, enabling seamless data exchange for QA evaluation, monitoring, and reporting.

Key Functions

1. Interaction Data Ingestion

- Retrieve interaction data from external systems:
 - CRM / Helpdesk platforms (e.g., Zoho, Zendesk, Salesforce)
 - Chat platforms (website, messaging channels)
 - Telephony systems (call metadata, recordings, or transcripts)
- Support:
 - Batch ingestion (historical data)
 - Near real-time ingestion (event-driven or streaming)

2. Data Mapping & Normalization

- Map incoming data to standardized QA evaluation structures
- Normalize fields such as:
 - Interaction metadata (timestamps, channel, agent ID)
 - Customer identifiers (if available)
 - Interaction content (chat, email, voice transcripts)
- Handle missing or inconsistent data formats
- Enrich interaction context using historical data (if available)

3. QA Evaluation Data Processing

- Pass interaction data to AI agents for:
 - Quality evaluation (CQC, Message Rating)
 - Sentiment and risk detection
- Generate:
 - QA scores
 - Flags (quality issues, compliance risks)
 - Priority classification

4. Insight Delivery & Output Integration

- Deliver outputs through existing operational channels:
 - Email (daily QA reports – MVP)
 - Messaging tools (e.g., Slack alerts)
 - Dashboards or exports (future phases)
- Provide:
 - Performance summaries
 - Flagged interaction alerts
 - Agent-level insights

5. Workflow Alignment & Synchronization

- Align QA evaluation outputs with existing QA and operational processes
- Enable:
 - Review workflows for flagged interactions (HITL)
 - Coaching and performance management actions
- Maintain synchronization between:

- Interaction data sources
- QA evaluation outputs
- Reporting and audit logs

Integration Principle

The integration layer enables the system to function as a **seamless QA and performance intelligence layer on top of existing infrastructure**, ensuring:

- No disruption to agent workflows
- No changes to existing customer interaction systems
- Scalable data ingestion and evaluation
- Consistent and traceable QA outputs across systems

Integration (MVP Scope):

- API-based integration with **Zoho CRM (preferred)**
- Integration with **chat platforms and telephony systems**
- Event-driven triggers for workflows (e.g., customer inquiry, delivery update)

Logging & Monitoring (Basic):

- Logs for all interactions (chat and voice)
- Capture of intents, actions, and escalation outcomes
- Enables traceability for review, debugging, and reporting
- Supports generation of daily operational reports

10. TESTING STRATEGY

This section defines how the system will be validated prior to release, including high-level testing requirements, scenarios, and acceptance criteria to ensure **accuracy, reliability, and usability of the AI-driven QA and performance monitoring solution**.

1. Unit Testing

- Validate individual agents and components (e.g., Content Quality Checker, AI Message Rating, Sentiment Analysis, Reporting Agent)
- Ensure correct processing of inputs (interaction data, transcripts) and generation of expected outputs (QA scores, flags, insights)

2. AI Evaluation Testing

- Test QA scoring outputs against predefined evaluation criteria and sample datasets
- Validate:
 - Consistency of scoring across similar interactions
 - Detection of quality issues (tone, compliance, resolution gaps)
- Ensure adherence to SOPs and QA frameworks

3. Integration Testing

- Validate data flow between:

- Input sources (chat platforms, email systems, voice transcripts)
- AI evaluation layer (QA scoring, sentiment detection)
- Output channels (email reports, alerts, dashboards)
- Ensure correct data mapping, formatting, and synchronization
- Verify non-intrusive integration with existing systems

4. User Acceptance Testing (UAT)

- QA teams and supervisors validate outputs using real or simulated interaction data
- Confirm:
 - Accuracy and usefulness of QA scores and insights
 - Relevance of flagged interactions
 - Usability of reports and alerts
- Validate alignment with QA workflows and processes

5. End-to-End Workflow Testing

Validate the full QA lifecycle:

1. Interaction data ingestion
2. QA evaluation and scoring
3. Sentiment and risk detection
4. Flagging and prioritization
5. HITL validation (if triggered)
6. Report generation and delivery

Ensure seamless orchestration with no data loss or processing gaps

6. Data Quality & Edge Case Testing

- Test handling of:
 - Incomplete or noisy transcripts
 - Ambiguous or multi-intent interactions
 - Low-quality or inconsistent data inputs
- Validate robustness of QA evaluation under edge scenarios

7. HITL & Validation Testing

- Validate HITL triggers:
 - Low-confidence evaluations
 - High-risk or compliance flags
 - Negative sentiment cases
- Ensure:
 - Accurate routing to QA reviewers
 - Ability to override and validate scores
 - Proper capture of feedback for continuous improvement

8. Reporting & Output Validation

- Validate accuracy of:
 - Daily QA summary reports

- Agent scorecards and performance insights
- Ensure consistency between:
 - Raw interaction data
 - QA evaluation outputs
 - Reported metrics

10.1 High-Level Test Scenarios & Conditions

Scenario 1: Automated Interaction QA Evaluation

Conditions:

- High-volume interaction dataset (chat, email, voice transcripts)
- Mix of:
 - High-quality vs low-quality interactions
 - Compliant vs non-compliant cases
- Incomplete or noisy transcripts

Expected Outcome:

- AI correctly assigns QA scores based on defined criteria
- Detects quality issues (tone, SOP adherence, resolution gaps)
- Flags high-risk and low-quality interactions
- Maintains consistent scoring across similar interactions

Scenario 2: Real-Time QA Monitoring & Alerting

Conditions:

- Live interaction streams (chat or voice transcripts)
- Presence of:
 - Negative sentiment
 - Compliance risks
 - Low-quality responses
- Multiple concurrent interactions

Expected Outcome:

- AI detects issues in near real-time (<1–3 seconds latency)
- Flags and classifies high-risk interactions
- Triggers alerts to supervisors (e.g., Slack/email)
- Maintains performance under concurrent load

Scenario 3: AI-Driven Performance Reporting

Conditions:

- Aggregated interaction dataset (daily volume)
- Mixed agent performance levels
- Historical data for trend analysis

Expected Outcome:

- System generates accurate:
 - QA summaries

- Agent scorecards
 - Performance insights
- Identifies:
 - Top performers
 - Agents requiring attention
 - Recurring quality issues

Scenario 4: HITL Validation & Escalation

Conditions:

- Low-confidence QA evaluations
- High-risk or compliance-related flags
- Negative sentiment interactions

Expected Outcome:

- System routes flagged interactions to QA reviewers
- Reviewers can:
 - Validate or override AI scores
- Feedback is captured for continuous improvement
- Escalation logic is applied correctly

Scenario 5: Data Quality & Edge Case Handling

Conditions:

- Missing or incomplete interaction data
- Ambiguous or unclear interaction context
- Mixed language or noisy transcripts

Expected Outcome:

- System handles incomplete or inconsistent data gracefully
- Flags interactions with insufficient evaluation confidence
- Maintains robustness across edge cases

Scenario 6: Integration & Data Flow Validation

Conditions:

- Data ingestion from:
 - CRM / Helpdesk systems
 - Chat platforms
 - Voice transcripts
- API delays or partial data availability

Expected Outcome:

- Data is correctly ingested and mapped
- No data loss across pipeline
- QA evaluation and reporting remain consistent
- System handles delays or failures with fallback mechanisms

Scenario 7: High Volume / Load Testing

Conditions:

- Simulated high load:
 - 1,000–5,000 interactions/day
 - 50–200 concurrent processing events (for real-time monitoring)
- Mixed channels and interaction types

Expected Outcome:

- System maintains performance and processing speed
- No significant degradation in QA evaluation accuracy
- Stable performance under peak load

General Acceptance Criteria

- 100% interaction evaluation coverage
- 60–70% reduction in manual QA effort
- <1–3 minutes (or seconds for real-time) issue detection
- 80–90% of high-risk interactions correctly flagged
- Consistent QA scoring aligned with defined criteria
- Accurate and timely report generation (daily)
- HITL validation workflows functioning correctly
- No critical failures across end-to-end QA pipeline